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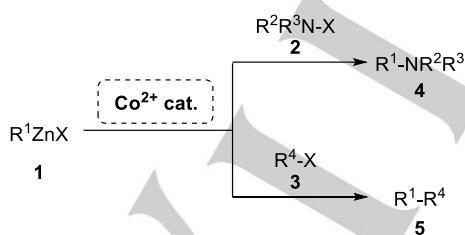
Cobalt-Catalyzed Cross-Couplings and Electrophilic Aminations using Organozinc Pivalates

Ferdinand H. Lutter^[a], Simon Graßl^[a], Lucie Grokenberger^[a], Maximilian S. Hofmayer^[a], Yi-Hung Chen^[b] and Paul Knochel^{[a]*}

Abstract: Transition metal catalyzed cross-couplings are essential methods in organic synthesis. Due to their comparable low toxicity and price, cobalt-salts offer a useful alternative to other common metal-catalysts (e.g. palladium-complexes). In this review, we report that cobalt-catalysts are especially well suited for cross-coupling reactions with organozinc pivalates, a class of organozinc reagents with enhanced stability towards air and moisture. Furthermore, we discuss applications of cobalt-catalysis in Negishi cross-coupling reactions (sp^2 - sp^2 , sp - sp^2 , sp - sp^3) as well as electrophilic aminations using organozinc reagents.

1. Introduction

Modern research towards novel targets for pharmacy and agriculture requires broadly applicable synthetic procedures.^[1] Within all the important transformations for this task, the formation of C-C and C-N bonds is of special interest. The most frequently used catalysts for cross-couplings as well as for aminations are palladium-complexes.^[2] Although palladium-catalysts provide a range of advantages such as a broad reaction scope, they often require expensive phosphine ligands and are costly by themselves. Nickel-catalysts represent a cheaper alternative for cross-couplings. However, an industrial usage is hindered by their high toxicity.^[3] The formation of C-N and C-C bonds using the reaction of organozinc reagents of type R^1ZnX **1** with electrophilic reagents such as R^2R^3N-X **2** and R^4-X **3** are effectively achieved by using cobalt-catalysis leading to valuable products of type **4** and **5**.



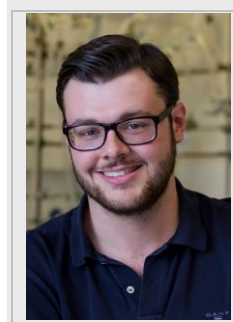
Scheme 1. Cobalt-catalyzed cross-couplings and aminations using organozinc reagents.

Cobalt-catalysis is of high interest, since cobalt salts have a low toxicity compared to nickel-catalysts.^[3] Furthermore, cobalt is about a 1000 times cheaper than palladium.^[4] These properties combined with an excellent catalytic activity have considerably increased the use of cobalt salts in organic synthesis.^[5] Whereas several cobalt-catalyzed reactions using organomagnesium reagents have been reported,^[6] it has recently been shown that organozinc reagents are also good reaction partners, since they tolerate a large number of functional groups.^[7] Other nucleophilic partners such as organomanganese^[8] reagents or organocopper^[9] species have also been used.

In this review, we will cover the recent advances in cobalt-catalyzed cross-couplings and aminations using organozinc reagents, emphasizing especially the use of organozinc pivalates. This class of organozinc reagents possesses an enhanced stability towards air and moisture and can be considerably stored as solids under argon at room temperature for several months.^[10] In fact, organozinc pivalates, abbreviated $RZnOPiv$, contain magnesium salts and would be more accurately represented as $(RZnCl \cdot Mg(OPiv)_2 \cdot LiCl)$.^[11] Therefore, first the main preparation methods of organozinc pivalates are described and then their utility for performing cobalt-catalyzed cross-couplings including diastereoselective cross-couplings and a convenient electrophilic amination reaction is demonstrated.

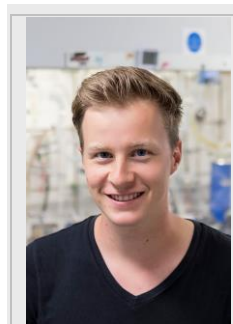
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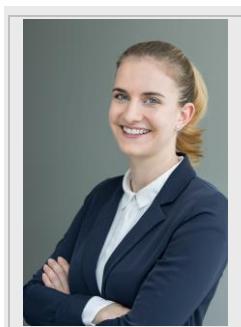
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on cobalt-catalyzed electrophilic aminations using zinc reagents.

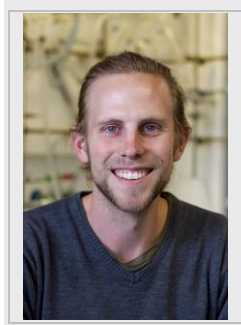
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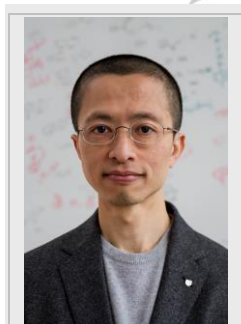
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Paul Knochel

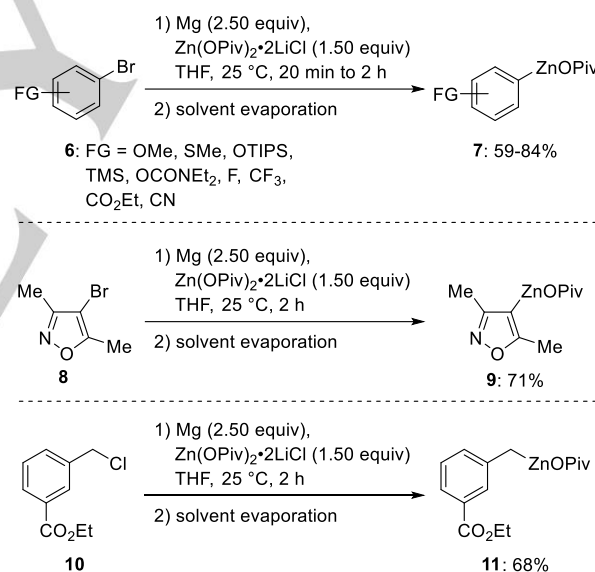
(1955) He studied at the University of Strasbourg (France) and completed his Ph.D. at the ETH-Zürich (D. Seebach). He spent four years at the University Pierre and Marie Curie in Paris (J.-F. Normant) and one year at Princeton University (M. F. Semmelhack). In 1987, he was appointed Professor at the University of Michigan. In 1992, he moved to the Philipps-Universität Marburg (Germany). In 1999, he then moved to the Chemistry Department of the Ludwig-Maximilians-Universität in Munich (Germany). His research interests include the development of novel organometallic reagents and methods for use in organic synthesis, asymmetric catalysis and natural



product synthesis. Prof. Knochel received many distinguished prizes including the Berthelot Medal of the Académie des Sciences (Paris), the Thieme-IUPAC Prize, the Otto Bayer Prize, the Leibniz-Prize, the Arthur C. Cope Scholar Award, Karl Ziegler Prize, the Nagoya Gold Medal, the H. C. Brown Award and the Paul Karrer Gold Medal. He is a member of the Académie des Sciences, the Bavarian Academy of Science, and the German Academy of Sciences Leopoldina. He is author of over 900 publications.

2. Preparation of organozinc pivalates

There are three main methods for preparing organozinc pivalates, bearing a broad range of functional groups.^[10] The first approach proceeds via a metal insertion and subsequent transmetalation. The starting materials are aryl and heteroaryl halides bearing functional groups.^[12] Thus, a range of functionalized unsaturated bromides **6** undergo an oxidative addition to magnesium turnings in the presence of $\text{Zn}(\text{OPiv})_2 \cdot 2\text{LiCl}$. The presence of this mild Lewis acid facilitates electron transfers from metallic magnesium to the organic substrates and provides the desired organozinc pivalates **7** within a few hours at 25 °C in high yields (Scheme 2).^[13]

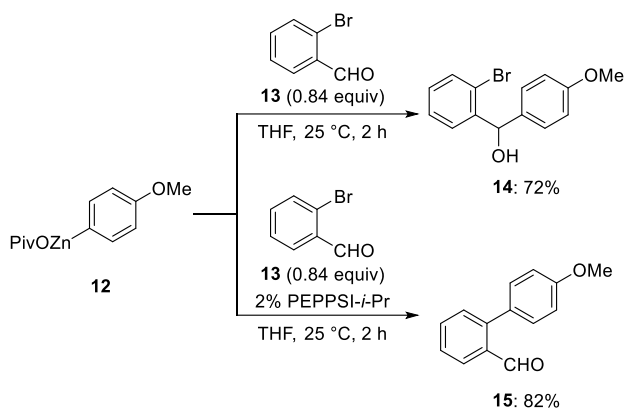


Scheme 2. Preparation of organozinc pivalates using magnesium turnings in the presence of $\text{Zn}(\text{OPiv})_2 \cdot 2\text{LiCl}$.

After solvent evaporation, solid powders are obtained with considerable air and moisture stability. This procedure can be extended to heterocyclic bromides such as bromoisoxazole **8**, which is converted to the solid organozinc pivalate **9** in 71% yield. Remarkably, benzylic chlorides such as **10** can be used in the same way with magnesium turnings in the presence of $\text{Zn}(\text{OPiv})_2 \cdot 2\text{LiCl}$, furnishing the benzylic zinc reagent **11** in 68% yield (Scheme 2).^[13] All these organozinc pivalates react well in Pd-catalyzed Negishi reactions and add to reactive carbonyl groups in the absence of a palladium catalyst. Thus, the arylzinc pivalate **12** undergoes a 1,2-addition to the bromobenzaldehyde

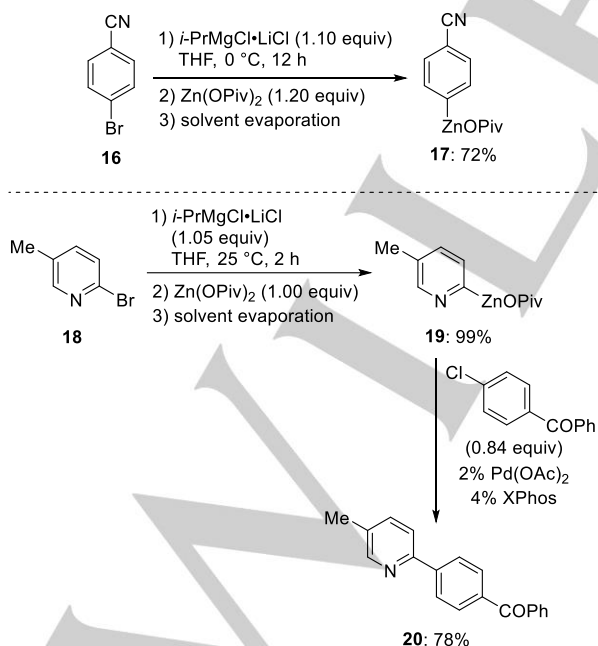
MINIREVIEW

13 in the absence of any catalyst, leading to the alcohol **14** in 72% yield whereas in the presence of Organ's catalyst (PEPPSI-*i*-Pr)^[14] the cross-coupling product **15** is obtained (Scheme 3).^[13]



Scheme 3. Reactivity of the solid arylzinc pivalate **12** with 2-bromobenzaldehyde **13**.

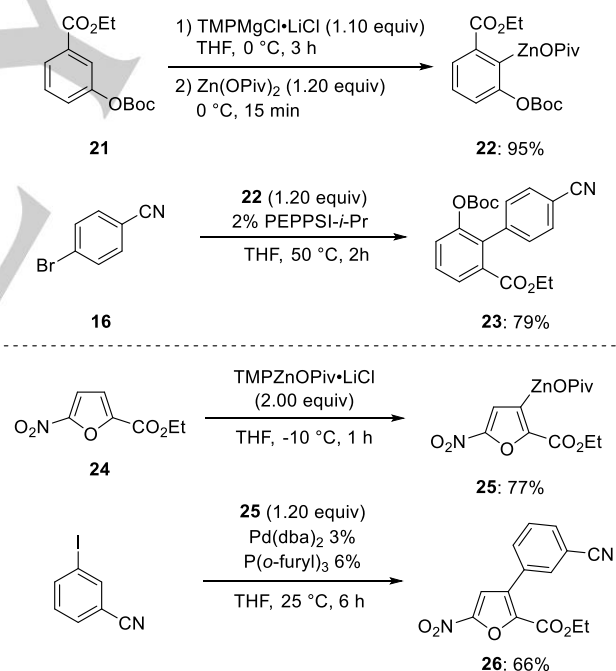
The second method for preparing aryl- and heteroarylzinc pivalates involves the use of the turbo-Grignard reagent *i*-PrMgCl•LiCl.^[15] Thus, bromobenzonitrile **16** was treated with *i*-PrMgCl•LiCl at 0 °C for 12 h furnishing after transmetalation with Zn(OPiv)₂ and subsequent solvent evaporation the solid arylzinc reagent **17** in 72% yield (Scheme 4).^[12] Similarly, various bromopyridines such as **18** readily react with *i*-PrMgCl•LiCl at 25 °C and providing after transmetalation and removal of the solvent the solid organozinc pivalate **19**, which smoothly couples with various aryl chlorides using Pd(OAc)₂ and Buchwald's ligand XPhos^[16] providing pyridines such as **20** (Scheme 4).^[17]



Scheme 4. Preparation of aryl- and heteroarylzinc pivalates using the turbo-Grignard *i*-PrMgCl•LiCl.

The method of choice for the generation of an organozinc pivalate reagent is highly depended on the corresponding (hetero)aryl halide. Thus, an insertion reaction proceeds smoothly for electron rich (hetero)aryl halides, whereas substrates bearing sensitive functional groups, such as ester, nitrile or amide moieties are best converted to the corresponding zinc pivalates using *i*-PrMgCl•LiCl and subsequent transmetalation.

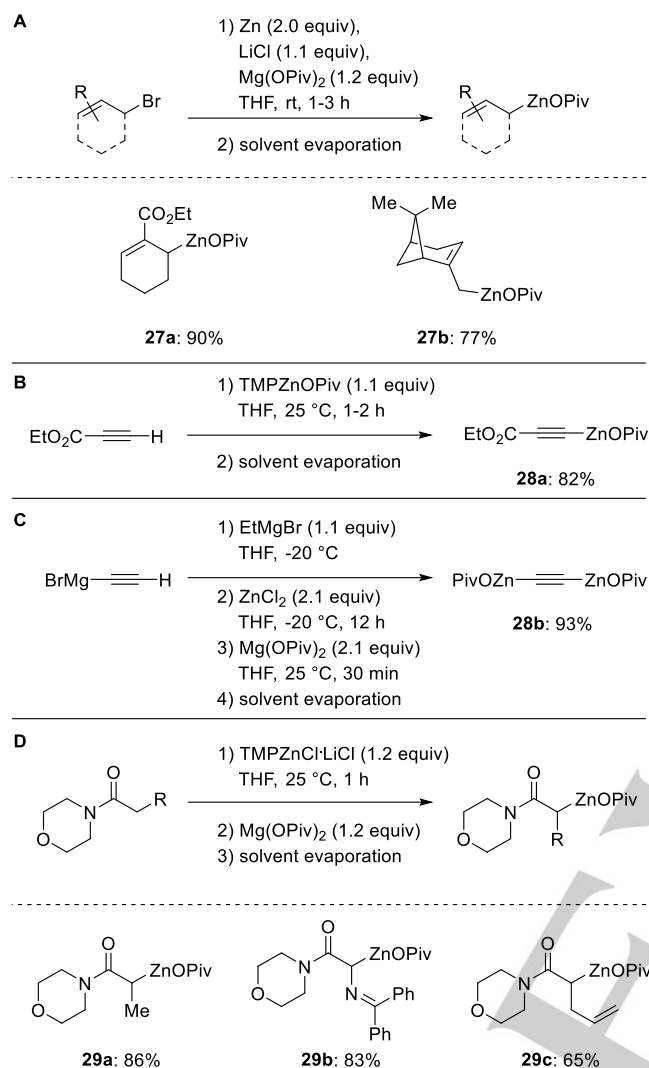
The third method involves a direct C-H metalation of arenes and heteroarenes using TMPMgCl•LiCl^[18] or TMPZnOPiv^[19]. The metalation reaction proceeds in the presence of directing groups, such as, esters, carbonates, aldehydes and halides. Thus, a range of polyfunctional arenes such as **21** are deprotonated with TMPMgCl•LiCl under mild conditions furnishing the corresponding magnesium derivative. After the addition of Zn(OPiv)₂ and solvent evaporation, the corresponding arylzinc pivalate **22** is obtained in 95% yield (Scheme 5).^[18] Pd-catalyzed cross-coupling with aryl bromide **16** using PEPPSI-*i*-Pr^[14] produces the biphenyl derivative **23** in 79% yield.^[18] If the substrate bears sensitive functional groups such as a nitro group, the usage of a milder base is required. Thus, the nitro-substituted furan derivative **24** reacts rapidly with TMPZnOPiv at -10 °C, providing the furylzinc pivalate **25**, which undergoes a Pd-catalyzed cross-coupling with various aryl iodides, leading to products such as **26** (Scheme 5).^[18-19]



Scheme 5. Preparation of aryl- and heteroarylzinc pivalates **22** and **25** using TMP-zinc and -magnesium bases.

These methods were extended to other classes of zinc reagents with enhanced air and moisture stability. Thus, it was possible to prepare functionalized allylic zinc pivalates such as **27a-b**^[20] (Scheme 6, A), as well as alkynylzinc pivalates, such as **28a**^[21] (Scheme 6, B) and **28b**^[22] (Scheme 6, C) or Reformatsky-reagents such as **29a-c**^[23] (Scheme 6, D).

MINIREVIEW

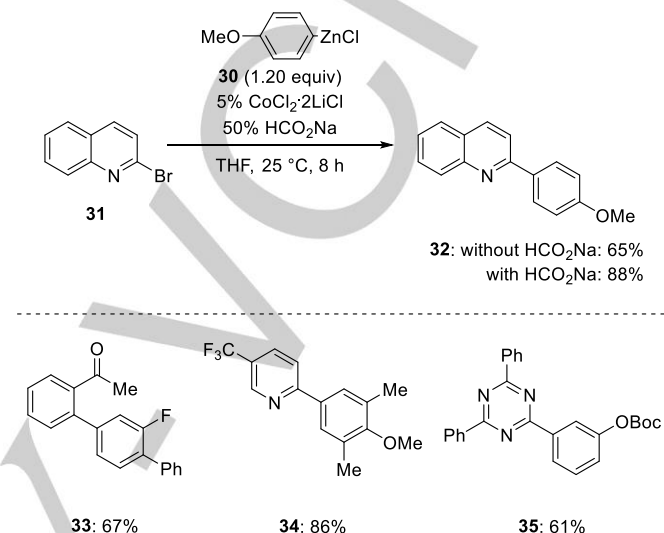


Scheme 6. Preparation of various functionalized allylic-, alkynyl- and Reformatsky-zinc pivalates.

3. Cobalt-catalyzed cross-couplings using organozinc pivalates

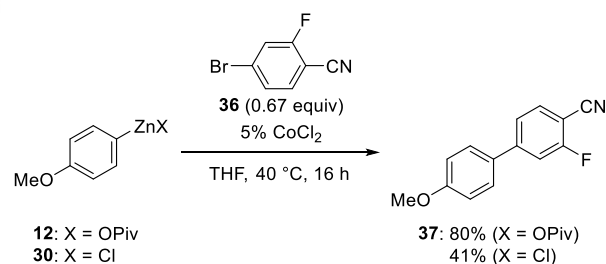
Although palladium-catalyzed cross-couplings still have the broadest reaction scope regardless of the nature of the electrophile or of the nucleophile, the price and the toxicity of this metal triggered research toward less toxic and less expensive catalysts. Cobalt-catalysis was found to be a viable alternative, although the scope of such cross-couplings is still too narrow to allow a very general application. Nevertheless, cobalt salts have been promising especially in combination with organozinc reagents as nucleophilic reaction partners. Thus, it was found that cobalt-catalysis enables the cross-coupling of various (hetero)arylzinc reagents with electron-deficient halides leading to the corresponding biaryls. The reaction outcome of these cross-couplings was significantly improved by the addition of 50% of a formate or pivalate salt (HCO₂Na or PivONa), which presumably

acts as a ligand for the cobalt catalyst thus making it more selective.^[26] Therefore, the reaction of *p*-anisylzinc chloride **30** with 2-bromoquinoline **31** leads to 65% of the arylated heterocycle **32**, whereas, in the presence of sodium formate the yield was increased to 88% (Scheme 7).^[26] However, the scope is limited to *ortho*-activated halides or electron deficient *N*-heterocycles leading to products such as **33–35** (Scheme 7).^[26]



Scheme 7. Cobalt-catalyzed cross-coupling using sodium formate as additive.

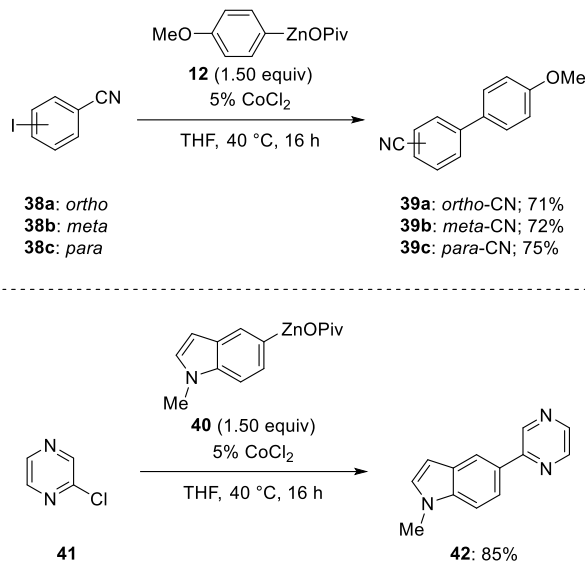
The beneficial effects of carboxylate ligands led to the use of aryl- and heteroarylzinc pivalates such as **12** for performing sp²-sp² cross-couplings instead of the corresponding arylzinc chloride **30** (Scheme 8).^[27] Thus, in the cross-coupling of the benzonitrile **36** with **12** or **30** a doubled yield was obtained by using arylzinc pivalate **12** compared to the arylzinc chloride **30**. This cross-coupling requires only 5% of the CoCl₂ catalyst and provides the product **37** in 80% yield (Scheme 8).^[27]



Scheme 8. Comparison of arylzinc pivalates and arylzinc chlorides in cobalt-catalyzed cross-couplings.

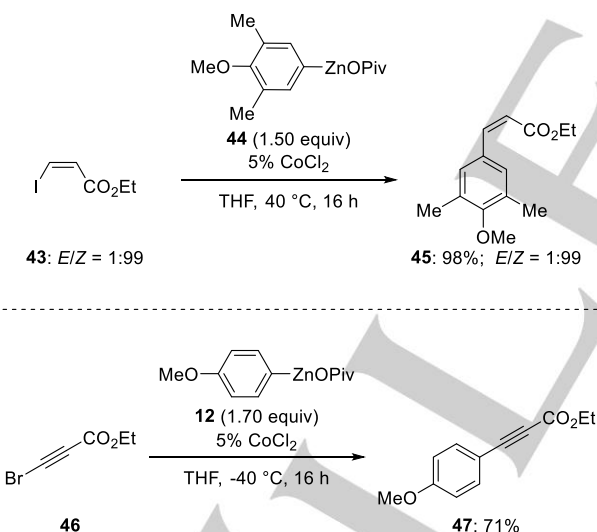
Furthermore, the use of arylzinc pivalates allows to significantly expand this cross-coupling reaction to all regioisomers of iodobenzonitrile **38a–c**, which react with *p*-anisylzinc pivalate (**12**) leading to the cyanobiphenyls **39a–c** in 71–75% yield (Scheme 9).^[27] Also, various heterocyclic building blocks can be coupled by this procedure. Thus, indolylzinc pivalate **40** reacts with 2-chloropyrazine (**41**) leading to the expected cross-coupling product **42** in 85% yield (Scheme 9).^[27]

MINIREVIEW



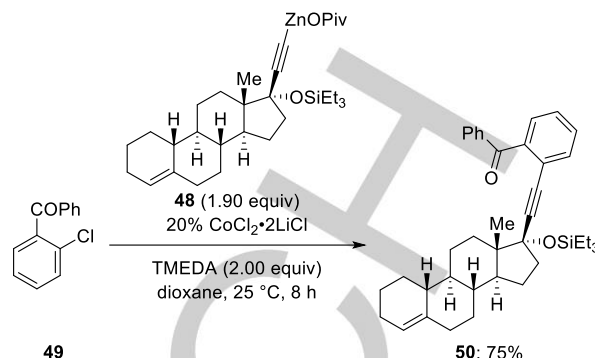
Scheme 9. Cobalt-catalyzed cross-couplings using aryl- and heteroarylzinc pivalates.

Both alkenyl iodides or bromides and alkynyl bromides undergo cobalt-catalyzed cross-couplings with arylzinc pivalates. Thus, the (*Z*)-iodoacrylate **43** couples with arylzinc pivalate **44** furnishing (*Z*)-acrylate **45** in 98% yield. Also, the alkynyl bromide **46** reacts with *p*-anisylzinc pivalate (**12**) leading to the alkynyl ester **47** in 71% yield (Scheme 10).^[27]



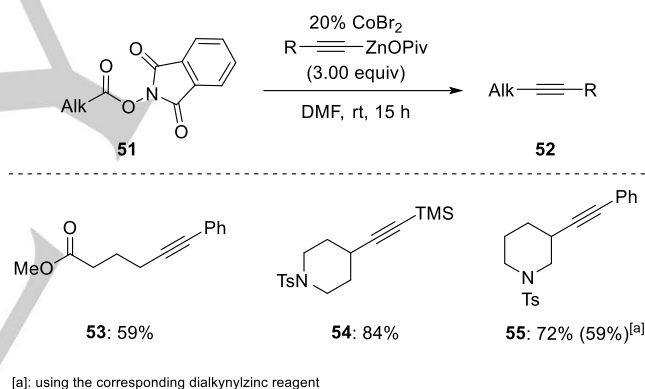
Scheme 10. Cobalt-catalyzed cross-couplings between alkenyl iodides and alkynyl bromides with arylzinc pivalates.

These robust cross-couplings^[28] can be extended to alkynylzinc pivalates. Thus, the steroid derived alkynylzinc pivalate **48** reacts with the chlorobenzophenone **49** providing the expected cross-coupling product **50** in 75% yield (Scheme 11).^[29a]



Scheme 11. Cobalt-catalyzed cross-couplings between alkynylzinc pivalate **49** and 2-chlorobenzophenone (**50**).

In 2018 Wang and co-workers described a cobalt-catalyzed decarboxylative Negishi cross-coupling of various functionalized alkyl redox-active ester with unsaturated organozinc reagents as coupling partners.^[29b] The coupling of alkynylzinc pivalates with the redoxactive esters of type **51** led to the coupling products **53–55** in 59–84% yield. Furthermore, the use of alkynylzinc pivalate reagents in comparison to dialkynyl zinc species in some cases led to enhanced yields (Scheme 12).



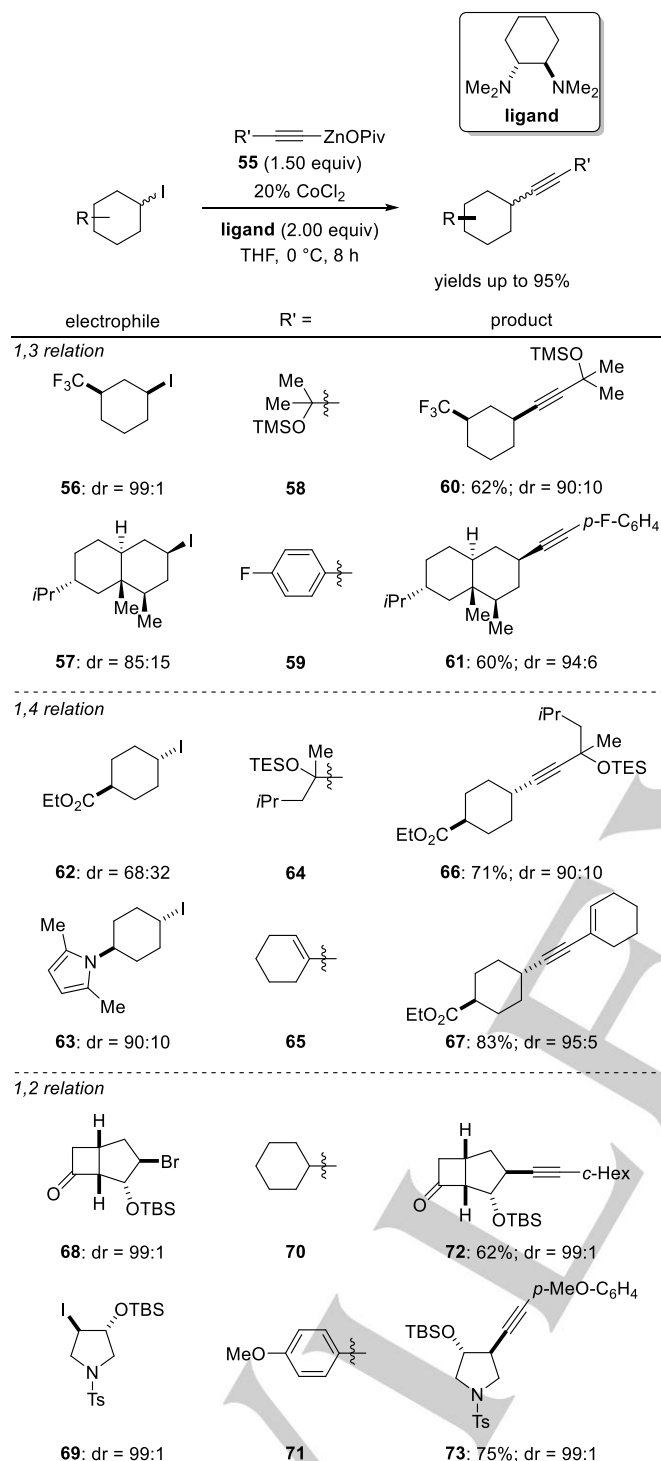
[a]: using the corresponding dialkynylzinc reagent

Scheme 12. Cobalt-catalyzed decarboxylative cross-coupling.

4. Cobalt-catalyzed diastereoselective cross-couplings of organozinc pivalates

Electron-poor arylmagnesium halides undergo diastereoselective cross-couplings with cyclic iodohydrins in the presence of iron salts.^[30] For extending the scope of this reaction to electron-rich arylmagnesium reagents, a cobalt catalysis proved to give excellent results.^[31] Alkynylzinc pivalates showed also a high propensity to undergo diastereoselective cross-couplings with various cycloalkyl iodides using a catalytic system consisting of 20% CoCl₂ and *trans*-tetramethylcyclohexane-1,2-diamine (2.00 equiv). In all cases, the formation of the more stable product occurs faster. Thus, various 1,3-substituted cyclohexyl iodides such as **56–57** undergo highly diastereoselective cross-couplings under the optimized reaction conditions with alkynylzinc pivalates **58–59** leading to the *syn*-1,3-disubstituted products **60–61** in 60–62% yield (Scheme 13).^[32]

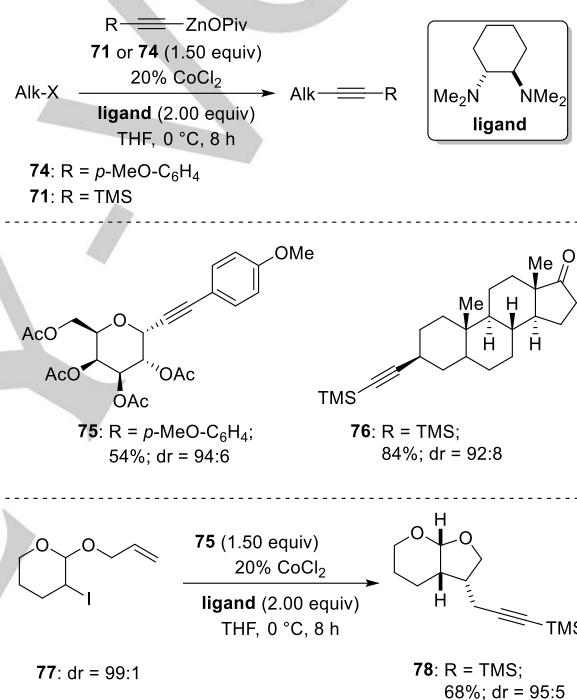
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Scheme 13. Diastereoselective cobalt-catalyzed cross-couplings between alkynylzinc pivalates with disubstituted iodocycloalkanes.

Also, 1,4-disubstituted iodocycloalkanes such as **62–63** undergo diastereoselective cross-couplings with various alkynylzinc pivalates **64–65** leading to the *anti*-1,4-disubstituted cyclohexane derivatives **66–67** in 71–83% yield. Various functional groups are compatible with these cross-couplings, including esters and

protected amino functionalities (Scheme 13).^[32] Finally, in the case of 1,2-disubstituted iodocyclohexanes, this cross-coupling provides a broad variety of structures. Thus, the cobalt-catalyzed cross-coupling of various iodo- or bromohydrins, such as **68–69** with alkynylzinc pivalates **70–71** in the presence of 20% CoCl₂ and *trans*-tetramethylcyclohexane-1,2-diamine (2.00 equiv) furnishes the *trans*-products **72–73** with excellent diastereoselectivities (Scheme 13).^[32] Furthermore, bromo-sugars or iodo-steroids can be diastereoselectively alkynylated with alkynylzinc pivalates **71** and **74** leading to the alkynes **75–76** (Scheme 14).^[32] The mechanism of these cobalt-catalyzed cross-couplings presumably proceeds via a radical pathway. This was shown by ring closure radical clock experiments of the cyclic iodide **77** with zinc reagent **75** leading to the cyclization product **78**.^[32,33]



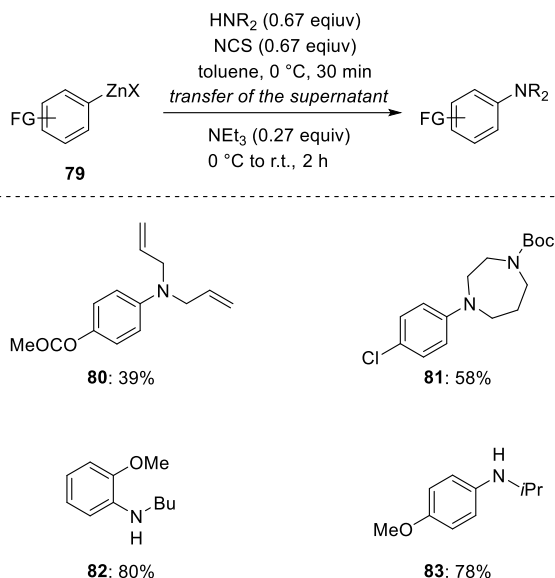
Scheme 14. Diastereoselective cobalt-catalyzed alkynylations of cyclic halides.

5. Cobalt-catalyzed electrophilic aminations with zinc organometallics

Nucleophilic aminations of aryl and heteroaryl halides catalyzed by palladium has given the access to numerous aryl and heteroaryl amines (Buchwald-Hartwig reaction).^[2b, 34] Alternatively, electrophilic aminations involving transition metal catalyst cross-coupling between main-group organometallics and electrophilic amination reagents such as *N*-hydroxylamine benzoates^[35] or chloroamines^[36] have attracted much attention since less expensive catalysts are usually required. For example, in his pioneering work, Johnson has described the electrophilic amination of organozinc derivatives with hydroxylamine benzoates using copper catalysis.^[35a] In 2013 Gosmini published a cobalt catalyzed electrophilic amination of organozinc halides using *N*-chloroamines.^[36g] The method was applied to prepare

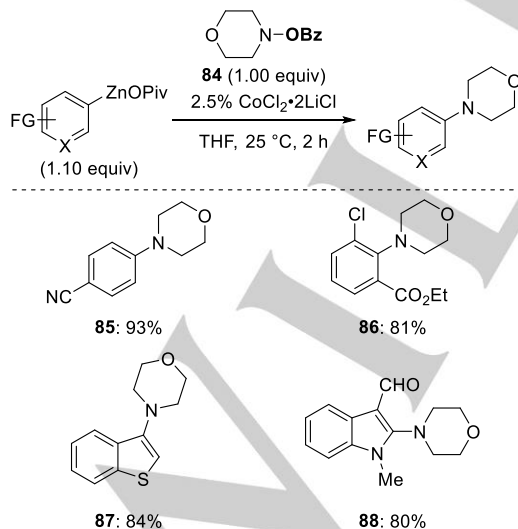
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secondary as well as tertiary amines. Using *N*-chlorosuccinimide the required chloroamines were prepared *in situ* and trapped with various organozinc reagents of type **79** leading to products **80–83** in 39–80% yield (Scheme 15). However, the usage of heterocyclic organometallics in combination with chloroamines remained a challenging task.



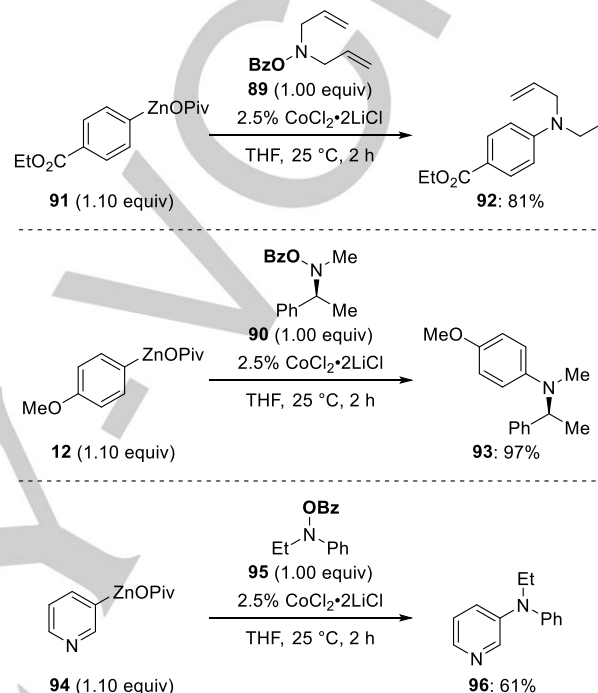
Scheme 15. Preparation of arylamines **80–83** via an electrophilic amination of organozinc halides with *in situ* prepared *N*-chloroamines.

Recently, we have shown that cobalt chloride complexed with two equivalents of lithium chloride is an excellent catalyst for the amination of various aryl- and heteroarylzinc pivalates with *N*-hydroxylamine benzoate **84** leading to polyfunctional amines such as **85–88** in high yields (Scheme 16).^[35k]



Scheme 16. Preparation of aryl- and heteroaryl amines **85–88** via an electrophilic amination of organozinc pivalates with the *N*-hydroxylamine benzoate **84**.

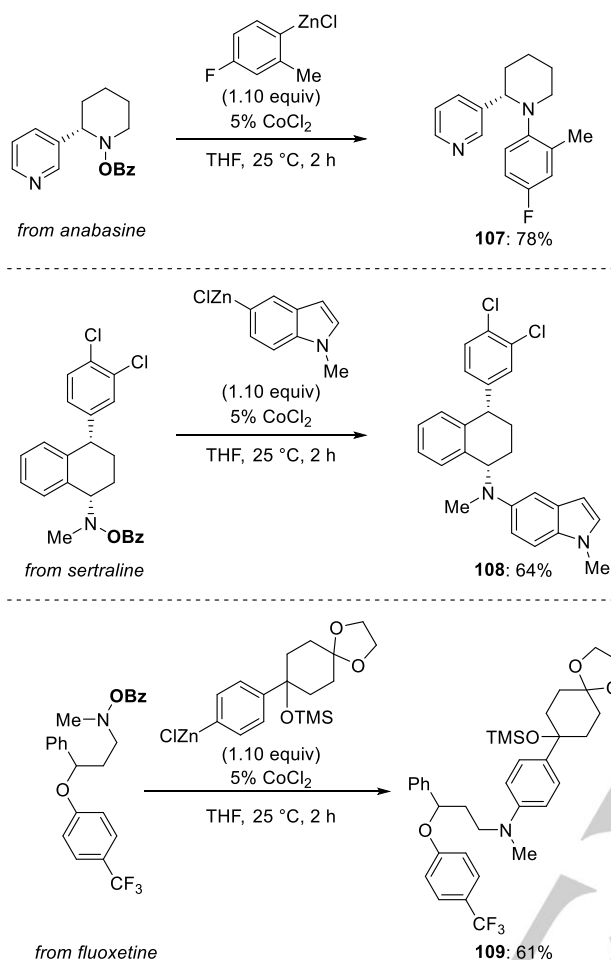
This electrophilic amination was readily extended to other *N*-hydroxylamine benzoates **89** and **90** available from the corresponding secondary amine via an oxidation with benzoyl peroxide.^[35k] Using the described reaction conditions the amination of arylzinc pivalates **91** and **12** led to the amines **92** and **93** in high yields. Also, 3-pyridylzinc pivalate (**94**) reacted with *N*-hydroxylamine benzoate **95** leading to the heterocyclic amine **96** in 61% yield (Scheme 17).^[35k]



Scheme 17. Preparation of polyfunctional amines via a cobalt-catalyzed electrophilic amination of organozinc pivalates with the *N*-hydroxylamine benzoates.

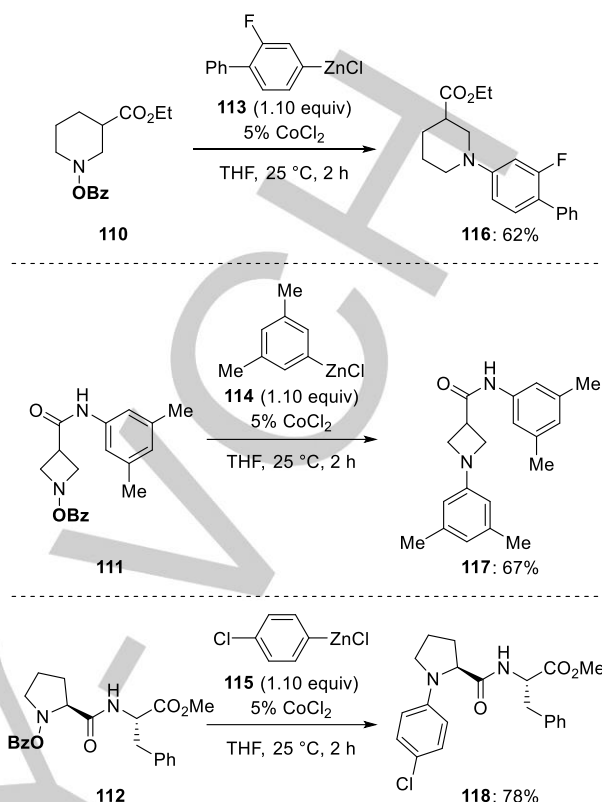
Using a procedure based on the work of O'Neill^[37] even complex amines and amino acids could be transformed into their corresponding hydroxylamine benzoates.^[38] All these *N*-hydroxylamine benzoates can be readily aminated using 5% CoCl₂ in THF within a reaction time of 2 h in good yields allowing a late stage amination of alkaloids like anabasine (see the preparation of **107**) or drugs like sertraline or fluoxetine (see the preparation of **108** and **109**, Scheme 18).^[38] A range of α - and β -amino-acids and a dipeptide can be converted into the

MINIREVIEW



Scheme 18. Cobalt-catalyzed amination using complex *N*-hydroxylamine benzoates.

N-hydroxylamine benzoates (**110**, **111** and **112**) and undergo selective amination reactions with arylzinc chlorides (**113**, **114** and **115**) leading to the arylated amino-acids and peptides (**116–118**) in 62–78% yield (Scheme 19). This method has also been applied to the synthesis of several drugs like penfluridol.^[38]



Scheme 19. Preparation of *N*-arylated amino-acids and a dipeptide using a cobalt-catalyzed electrophilic amination.

6. Conclusion

The need for convenient and economically sustainable transition metal-catalyzed cross-couplings was a main issue in academic as well as in industrial research during the last decades. Cobalt salts showed to be a versatile alternative to other frequently used transition-metal catalysts (e.g. palladium or nickel). These catalysts combine a relative low toxicity and price with a high catalytic activity in cross-coupling reactions. Thus, cobalt catalysis was applied to several Negishi cross-couplings including sp²-sp², sp-sp², sp-sp³. Even diastereoselective versions have been successfully developed. Furthermore, cobalt-catalyzed electrophilic aminations of organozinc reagents using *N*-hydroxylamine benzoates have been reported. This C-N bond forming protocol enables the functionalization of complex secondary amines such as the drug Fluoxetine. These transformations could be improved in many cases by using organozinc pivalates, a new class of zinc organometallic reagents that can be stored as solids under argon for months. Thus, the combination of cobalt-catalysis and organozinc pivalates could be a valuable tool in organic synthesis.

MINIREVIEW

Acknowledgements

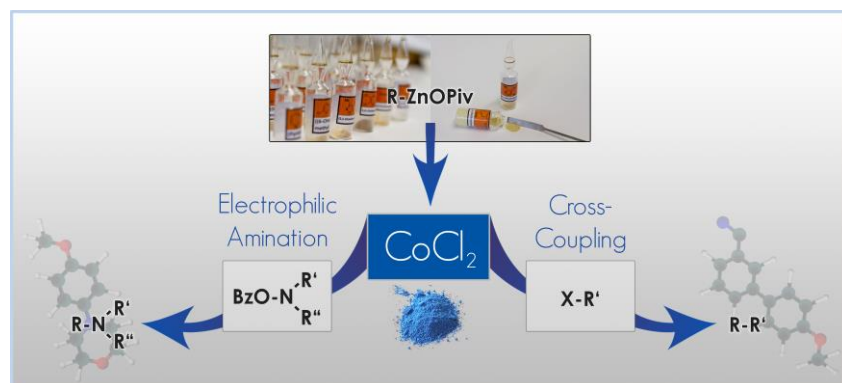
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Keywords: cobalt • cross-coupling • amination • C-C bond formation • organometallic reagent

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MINIREVIEW

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Page No. – Page No.

Cobalt-Catalyzed Cross-Couplings and Electrophilic Aminations using Organozinc Pivalates

Transition metal catalyzed C-C and C-N bond forming reactions are of great interest for the elaboration of pharmaceuticals and agrochemicals. Due to their low price and toxicity, cobalt-salts offer a great alternative to other common metal-catalyst (e.g. palladium-complexes). Furthermore, cobalt-catalysis is well compatible with organozinc pivalates, a class of organozinc reagents with enhanced stability towards air and moisture. In this review, the versatile applications of cobalt-catalysis in Negishi cross-coupling reactions and electrophilic aminations using organozinc pivalates are discussed.